

Question: Let R consists of set of all points 'z' such that $0 < |z| \leq 1$ and $f(z) = z^2$. Verify that function is uniformly continuous.

Solution: Given that $f(z) = z^2$, $0 < |z| \leq 1$ and $z_1, z_2 \in R$ then

$$0 < |z_1| \leq 1, \quad 0 < |z_2| \leq 1$$

$$\text{Consider } |f(z_1) - f(z_2)| = |z_1^2 - z_2^2| = |(z_1 - z_2)(z_1 + z_2)| = |z_1 - z_2||z_1 + z_2|$$

Since $0 < |z_1| \leq 1$, $0 < |z_2| \leq 1$ therefor

$$|f(z_1) - f(z_2)| \leq 2|z_1 - z_2| \quad \because |z_1 + z_2| = |1 + 1| = 2$$

$$|f(z_1) - f(z_2)| \leq 2\delta \quad \text{Whenever } |z_1 - z_2| < \delta = \frac{\epsilon}{2}$$

$$|f(z_1) - f(z_2)| < 2 \cdot \frac{\epsilon}{2} \quad \text{Whenever } |z_1 - z_2| < \delta = \frac{\epsilon}{2}$$

$$|f(z_1) - f(z_2)| < \epsilon \quad \text{Whenever } |z_1 - z_2| < \delta$$

$f(z) = z^2$ is uniformly continuous on R

Note that in expression $|f(z_1) - f(z_2)| \leq 2|z_1 - z_2|$ equality will hold when $|z_1| = 1, |z_2| = 1$.

Exercise 16: Show that given functions are uniformly continuous at $z = z_0$ or not?

- $f(z) = \frac{1}{z}$ in the region $|z| < 1$ (not continuous)
- $f(z) = z^2 - 1$ in the region $|z| \leq 4$

Theorem: A composition of continuous functions is itself continuous.

Proof: Suppose that $g[f(z)]$ be a continuous function defined for all $f(z)$ in the Neighborhood of $f(z_0)$ then by definition;

$$|g[f(z)] - g[f(z_0)]| < \epsilon \quad \text{whenever} \quad |f(z) - f(z_0)| < \gamma \dots\dots(i)$$

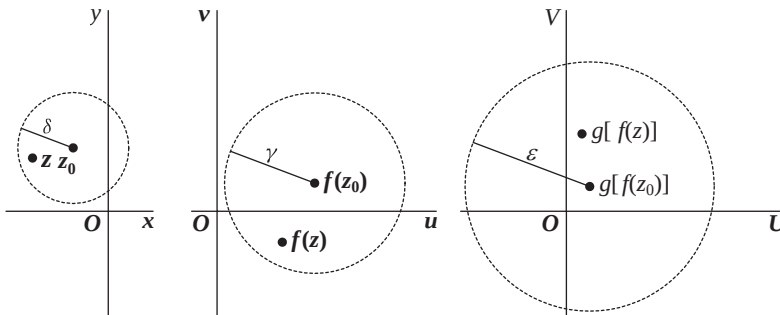
Similarly suppose that $f(z)$ be a continuous function defined for all z in the Neighborhood of z_0 then by definition;

$$|f(z) - f(z_0)| < \gamma \quad \text{whenever} \quad |z - z_0| < \delta \dots\dots(ii)$$

Combining (i) and (ii)

$$|g[f(z)] - g[f(z_0)]| < \epsilon \quad \text{whenever} \quad |z - z_0| < \delta$$

Hence composition of continuous functions is itself continuous.



Theorem: If a function $f(z)$ is continuous and nonzero at a point z_0 ,

Then $f(z) \neq 0$ throughout some neighborhood of that point.

Proof: Suppose that function $f(z)$ is continuous and nonzero at a point z_0 , and $f(z) = 0$ throughout some neighborhood of that point. Then by definition of continuity

$$|f(z) - f(z_0)| < \epsilon \quad \text{whenever} \quad |z - z_0| < \delta$$

take $\epsilon = \frac{|f(z_0)|}{2}$ and $f(z) = 0$ then

$$|f(z_0)| < \frac{|f(z_0)|}{2} \quad \text{whenever} \quad |z - z_0| < \delta \quad \text{Which is contradiction.}$$

Hence If a function $f(z)$ is continuous and non-zero at a point z_0 ,

Then $f(z) \neq 0$ throughout some neighborhood of that point.

Theorem: If a function $f(z)$ is continuous throughout a region R that is Both closed and bounded. Then there exist a non-negative real number M such that $|f(z)| \leq M$ for all point ' z ' in R .

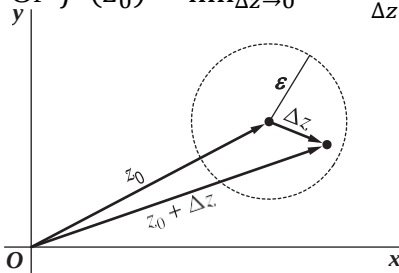
Proof: Suppose that function $f(z) = u(x, y) + iv(x, y)$ is continuous. Then $|f(z)|$ Will be continuous throughout R and thus reaches a maximum value (say) M somewhere in R . then $|f(z)| \leq M$ for all point ' z ' in R . And we will say that f is bounded on R .

DERIVATIVES:

Given $W = f(z)$ be a single valued function defined in a domain D and let z_0 be any fixed point in D . Then $W = f(z)$ is said to have a derivative at z_0 if the following limits exists.

$$\frac{df}{dz} = \frac{dW}{dz} = f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

$$\text{Or } f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \quad \text{where } \Delta z = z - z_0$$



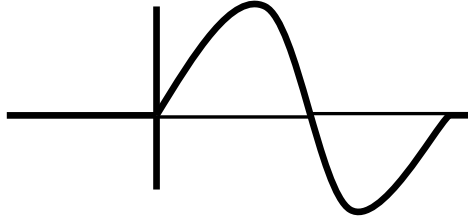
ϵ, δ FORM

A complex valued function $W = f(z)$ is said to be a derivative at $z = z_0$ if for each positive number ϵ , there is a positive number δ such that

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \epsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta.$$

Remark:

- i. Instantaneous rate of change of one variable with respect to other Variable is called **derivative** and method to find derivative is called Differentiation or Differentiability.
- ii. **Graphically**, the derivative of a function corresponds to the slope of its tangent line at one specific point.



- iii. Since $W = f(z)$ defined through a neighborhood of 'z' therefore the Number $f(z_0 + \Delta z)$ is always defined for $|\Delta z|$ sufficiently small.

Theorem: Prove that $W = f(z)$ is differentiable then it will be continuous.

Proof: Suppose that $W = f(z)$ is differentiable at a point z_0 then

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Now we have to prove that $f(z)$ is continuous.

$$\text{Consider } f(z) - f(z_0) = \frac{f(z) - f(z_0)}{z - z_0} \times z - z_0$$

$$\text{Then } \lim_{z \rightarrow z_0} [f(z) - f(z_0)] = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \times \lim_{z \rightarrow z_0} z - z_0$$

$$[\lim_{z \rightarrow z_0} f(z)] - f(z_0) = f'(z_0) \times 0 \Rightarrow [\lim_{z \rightarrow z_0} f(z)] - f(z_0) = 0$$

$$\Rightarrow \lim_{z \rightarrow z_0} f(z) = f(z_0) \Rightarrow W = f(z) \text{ is continuous}$$

Remark: Convers of above theorem is not true.

Question: Show that $f(z) = \bar{z}$ is continuous but not differentiable.

Solution: let $z_0 \in \mathbb{C}$ then $z_0 = x_0 + iy_0$ and given $f(z) = \bar{z}$ then

- i. $f(z_0) = \bar{z}_0 = \overline{x_0 + iy_0} = x_0 - iy_0$. i.e. $f(z)$ is defined at $z = z_0$
- ii. $\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow (x_0 + iy_0)} \bar{z} = \overline{x_0 + iy_0} = x_0 - iy_0$
- iii. The value and limit agree at $z = z_0$ i.e. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

Hence $f(z) = \bar{z}$ is continuous on $\mathbb{C}$.

Now for **Differentiability**.

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{z \rightarrow z_0} \frac{\bar{z} - \bar{z}_0}{z - z_0} = \lim_{z \rightarrow z_0} \frac{\overline{z - z_0}}{z - z_0}$$

Let $z - z_0 = \Delta z$ then if $z \rightarrow z_0$ then $\Delta z \rightarrow 0$

$$\Rightarrow f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z} = \pm 1 \text{ i.e. when '}\Delta z\text{' is real then limit is '1' and}$$

when ' Δz ' is imaginary then limit is ' -1 ' therefore

limit does not exist and function is not differentiable.

REMARK: At $z = 0$ the function is continuous as well as differentiable.

Question: Show that $f(z) = |z|^2 = z\bar{z}$ is continuous but not differentiable.

Solution: let $z_0 \in \mathbb{C}$ and given $f(z) = |z|^2$ then

- $f(z_0) = |z_0|^2$. i.e. $f(z)$ is defined at $z = z_0$
- $\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} |z|^2 = |z_0|^2$
- The value and limit agree at $z = z_0$ i.e. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

Hence $f(z) = |z|^2$ is continuous on $\mathbb{C}$.

Now for **Differentiability**.

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{z \rightarrow z_0} \frac{|z|^2 - |z_0|^2}{z - z_0} = \lim_{z \rightarrow z_0} \frac{z\bar{z} - z_0\bar{z}_0}{z - z_0}$$

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{z\bar{z} - \bar{z}z_0 + \bar{z}z_0 - z_0\bar{z}_0}{z - z_0} = \lim_{z \rightarrow z_0} \frac{\bar{z}(z - z_0) + z_0(\bar{z} - \bar{z}_0)}{z - z_0} = \lim_{z \rightarrow z_0} \left[\bar{z} + \frac{z_0(\bar{z} - \bar{z}_0)}{z - z_0} \right]$$

Let $z - z_0 = \Delta z$ then if $z \rightarrow z_0$ then $\Delta z \rightarrow 0$

$$\Rightarrow f'(z_0) = \lim_{\Delta z \rightarrow 0} \left[\bar{z} + \frac{z_0(\Delta \bar{z})}{\Delta z} \right]$$

when ' Δz ' is real then limit is ' $\bar{z} + z_0$ ' and

when ' Δz ' is imaginary then limit is ' $\bar{z} - z_0$ '

Therefore limit does not exist and function is not differentiable.

REMARK: At $z = 0$ the function is continuous as well as differentiable.

Question: Show that $f'(z) = -\frac{1}{z^2}$ when $f(z) = \frac{1}{z}$ at each $z \neq 0$

Solution: by using the definition

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \left[\frac{1}{(z + \Delta z)} - \frac{1}{z} \right] \cdot \frac{1}{\Delta z} = \lim_{\Delta z \rightarrow 0} \left[\frac{-1}{(z + \Delta z)z} \right]$$

$$\Rightarrow f'(z) = -\frac{1}{z^2} \text{ at each } z \neq 0$$

Exercise 17: (visit @ Youtube "learning with Usman Hamid")

i. Show that $f'(z) = 2z$ when $f(z) = z^2$ at each $z \neq 0$

ii. Find $f'(z)$ when

a) $f(z) = \left(\frac{1}{3}z^3 - \frac{1}{3z^3} \right)^4$

b) $f(z) = \frac{1}{(z^4 + 2)^2}$

c) $f(z) = 3z^2 - 2z + 4$

d) $f(z) = (2z^2 + i)^5$

e) $f(z) = \frac{z+1}{2z+1} ; z \neq -\frac{1}{2}$

f) $f(z) = \frac{(1+z^2)^4}{z^2} ; z \neq 0$

iii. Suppose that $f(z_0) = g(z_0) = 0$ and $f'(z_0)$ and $g'(z_0)$ exist, where

$$g'(z_0) \neq 0 \text{ then show that } \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$$

iv. Show that $f'(0)$ does not exist when $f(z) = \begin{cases} \frac{\bar{z}^2}{z} & z \neq 0 \\ 0 & z = 0 \end{cases}$

v. Show that $f'(z)$ does not exist at any point ' z ' when

a) $f(z) = \operatorname{Re} z$

c) $f(z) = |z|$

b) $f(z) = \operatorname{Im} z$

d) $f(z) = \arg(z)$

CHAPTER 2

ANALYTIC FUNCTIONS

ANALYTIC / REGULAR / HOLOMORPHIC FUNCTION

A complex valued function $W = f(z)$ is said to be Analytic in domain D if

- i. $W = f(z)$ is single valued.
- ii. $W = f(z)$ is differentiable in domain D.

CAUCHY RIEMANN EQUATIONS

Let $f(z) = u + iv$ be a complex valued function, whose first order partial derivative exists, then following pair of equations is called CR equation;

$$u_x = v_y \text{ and } u_y = -v_x$$

We may also write $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

CAUCHY-RIEMANN EQUATIONS IN RECTANGULAR COORDINATES (necessary condition)

Suppose that $f(z) = u(x, y) + iv(x, y)$ and that $f'(z)$ exists at a point z_0

Then the first-order partial derivatives of u and v must exist at (x_0, y_0) ,

Then they must satisfy the Cauchy-Riemann equations i.e. $u_x = v_y$ and $u_y = -v_x$

Proof: Given that $f'(z)$ exists at a point z_0 $f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$

We have to prove $u_x = v_y$ and $u_y = -v_x$

$$\begin{aligned} \text{Consider } f'(z_0) &= \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \\ \Rightarrow f'(z_0) &= \lim_{\Delta z \rightarrow 0} \frac{[u(x_0 + \Delta x, y_0 + \Delta y) + iv(x_0 + \Delta x, y_0 + \Delta y)] - [u(x_0, y_0) + iv(x_0, y_0)]}{\Delta x + i\Delta y} \end{aligned}$$

Along horizontal axis: Take $\Delta x \rightarrow 0$ and $\Delta y = 0$

$$\begin{aligned} \Rightarrow f'(z_0) &= \lim_{\Delta x \rightarrow 0} \frac{[u(x_0 + \Delta x, y_0) + iv(x_0 + \Delta x, y_0)] - [u(x_0, y_0) + iv(x_0, y_0)]}{\Delta x} \\ \Rightarrow f'(z_0) &= \lim_{\Delta x \rightarrow 0} \frac{[u(x_0 + \Delta x, y_0) - u(x_0, y_0)] + i[v(x_0 + \Delta x, y_0) - v(x_0, y_0)]}{\Delta x} \\ \Rightarrow f'(z_0) &= \lim_{\Delta x \rightarrow 0} \frac{[u(x_0 + \Delta x, y_0) - u(x_0, y_0)]}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{[v(x_0 + \Delta x, y_0) - v(x_0, y_0)]}{\Delta x} \\ \Rightarrow f'(z_0) &= u_x(x_0, y_0) + iv_x(x_0, y_0) \dots \dots \dots (i) \end{aligned}$$

Along vertical axis: Take $\Delta y \rightarrow 0$ and $\Delta x = 0$

$$\Rightarrow f'(z_0) = \lim_{\Delta y \rightarrow 0} \frac{[u(x_0, y_0 + \Delta y) + iv(x_0, y_0 + \Delta y)] - [u(x_0, y_0) + iv(x_0, y_0)]}{i\Delta y}$$

$$\Rightarrow f'(z_0) = \lim_{\Delta y \rightarrow 0} \frac{[u(x_0, y_0 + \Delta y) - u(x_0, y_0)] + i[v(x_0, y_0 + \Delta y) - v(x_0, y_0)]}{i\Delta y}$$

$$\Rightarrow f'(z_0) = \lim_{\Delta y \rightarrow 0} \frac{[u(x_0, y_0 + \Delta y) - u(x_0, y_0)]}{i\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{[v(x_0, y_0 + \Delta y) - v(x_0, y_0)]}{i\Delta y}$$

$$\Rightarrow f'(z_0) = -i \lim_{\Delta y \rightarrow 0} \frac{[u(x_0, y_0 + \Delta y) - u(x_0, y_0)]}{\Delta y} + \lim_{\Delta y \rightarrow 0} \frac{[v(x_0, y_0 + \Delta y) - v(x_0, y_0)]}{\Delta y}$$

$$\Rightarrow f'(z_0) = -iu_y(x_0, y_0) + v_y(x_0, y_0) = v_y - iu_y \quad \dots \dots \dots (ii)$$

Comparing real and imaginary parts of (i) and (ii) we get

$$u_x = v_y \quad \text{and} \quad v_x = -u_y \quad \text{which are CR equations.}$$

$$\text{We may also write } u_x = v_y \quad \text{and} \quad u_y = -v_x$$

Theorem (sufficient condition)

Suppose that $f(z) = u(x, y) + iv(x, y)$ is defined throughout some neighborhood Of a point z_0 and the first-order partial derivatives of u and v must exist at (x_0, y_0) , Also Cauchy–Riemann equations $u_x = v_y$ and $u_y = -v_x$ hold Then $f'(z)$ exists at a point z_0

Proof:

Suppose that CR equations holds i.e. $u_x = v_y$ and $u_y = -v_x$ and continuous

Consider

$$f(z_0 + \Delta z) - f(z_0) =$$

$$[u(x_0 + \Delta x, y_0 + \Delta y) + iv(x_0 + \Delta x, y_0 + \Delta y)] - [u(x_0, y_0) + iv(x_0, y_0)]$$

Adding and subtracting $u(x_0 + \Delta x, y_0)$ and $v(x_0, y_0 + \Delta y)$

$$f(z_0 + \Delta z) - f(z_0) =$$

$$[u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0 + \Delta x, y_0) + u(x_0 + \Delta x, y_0) - u(x_0, y_0)]$$

$$+ i[v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0 + \Delta y) + v(x_0, y_0 + \Delta y) - v(x_0, y_0)]$$

Since

$$u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0 + \Delta x, y_0) = \Delta y u_y + \epsilon_1 \Delta y$$

$$u(x_0 + \Delta x, y_0) - u(x_0, y_0) = \Delta x u_x + \epsilon_2 \Delta x$$

$$v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0 + \Delta y) = \Delta x v_x + \epsilon_3 \Delta x$$

$$v(x_0, y_0 + \Delta y) - v(x_0, y_0) = \Delta y v_y + \epsilon_4 \Delta y$$

Then $f(z_0 + \Delta z) - f(z_0) =$

$$[\Delta y u_y + \epsilon_1 \Delta y + \Delta x u_x + \epsilon_2 \Delta x] + i[\Delta x v_x + \epsilon_3 \Delta x + \Delta y v_y + \epsilon_4 \Delta y]$$

using CR equations i.e. $u_x = v_y$ and $u_y = -v_x$

Then $f(z_0 + \Delta z) - f(z_0) =$

$$[-\Delta y v_x + \epsilon_1 \Delta y + \Delta x u_x + \epsilon_2 \Delta x] + i[\Delta x v_x + \epsilon_3 \Delta x + \Delta y u_y + \epsilon_4 \Delta y]$$

$$= [u_x(\Delta x + i\Delta y) + iv_x(\Delta x + i\Delta y) + \Delta x(\epsilon_2 + i\epsilon_3) + \Delta y(\epsilon_1 + i\epsilon_4)]$$

$$= [u_x(\Delta x + i\Delta y) + iv_x(\Delta x + i\Delta y) + \Delta x\delta_1 + \Delta y\delta_2]$$

Dividing by $\Delta z = \Delta x + i\Delta y$

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{[u_x(\Delta x + i\Delta y) + iv_x(\Delta x + i\Delta y) + \Delta x\delta_1 + \Delta y\delta_2]}{\Delta x + i\Delta y}$$

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{[u_x + iv_x](\Delta x + i\Delta y) + \Delta x\delta_1 + \Delta y\delta_2}{\Delta x + i\Delta y}$$

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = (u_x + iv_x) + \frac{\Delta x\delta_1 + \Delta y\delta_2}{\Delta x + i\Delta y}$$

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - (u_x + iv_x) = \frac{\Delta x\delta_1 + \Delta y\delta_2}{\Delta x + i\Delta y}$$

$$\left| \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - (u_x + iv_x) \right| = \left| \frac{\Delta x\delta_1 + \Delta y\delta_2}{\Delta x + i\Delta y} \right| \leq \delta_1 \frac{|\Delta x|}{|\Delta x + i\Delta y|} + \delta_2 \frac{|\Delta y|}{|\Delta x + i\Delta y|}$$

$$\left| \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - (u_x + iv_x) \right| < \delta_1 + \delta_2 < \epsilon \quad \therefore \frac{|\Delta x|}{|\Delta x + i\Delta y|} < 1 \text{ also } \frac{|\Delta y|}{|\Delta x + i\Delta y|} < 1$$

$$\Rightarrow f'(z_0) = u_x + iv_x \text{ or } \Rightarrow f'(z_0) = u_x - iu_y \text{ or } \Rightarrow f'(z_0) = v_y - iu_y$$

This is the definition of differentiability.

Remark:

- i. If $W = f(z)$ is a non – constant real valued function (say) $f(z) = |z|^2$
Then CR equations do not hold.
- ii. If $W = f(z)$ is a constant real valued function then CR equations hold.
- iii. If $W = f(\bar{z})$ then CR equations do not hold.
- iv. If a function is analytic, CR equations necessarily hold, but if CR equations are satisfied for a function then the function may or may not be analytic.
- v. If a function involves $\bar{z}$ then without verifying CR equations we can Say the function is non – analytic.
- vi. We may use $u_x(0,0) = \lim_{x \rightarrow 0} \frac{u(x,0) - u(0,0)}{x}$ at origin
Instead of $u_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{[u(x + \Delta x, 0) - u(0,0)]}{\Delta x}$ and vice versa.

Example: Since the function $f(z) = z^2 = x^2 - y^2 + i2xy$

is differentiable everywhere and that $f'(z) = 2z$. Verify that the CR equations are satisfied everywhere.

Solution: write $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

Thus $u_x = 2x = v_y$, $u_y = -2y = -v_x$.

Moreover, $f'(z) = 2x + i2y = 2(x + iy) = 2z$. i.e. $f'(z)$ exists.

Example: Prove that for the function $f(z) = \begin{cases} \frac{x^3(1+i)-y^3(1-i)}{x^2+y^2} & z \neq 0 \\ 0 & z = 0 \end{cases}$

CR equations are satisfied at $z = 0$ but function is not differentiable i.e. not analytic at $z = 0$

Solution: Given that $f(z) = \frac{x^3(1+i)-y^3(1-i)}{x^2+y^2} = \frac{(x^3-y^3)+i(x^3+y^3)}{x^2+y^2}$

Then $u(x, y) = \frac{x^3-y^3}{x^2+y^2}$ also $v(x, y) = \frac{x^3+y^3}{x^2+y^2}$

Then at origin

$$\Rightarrow u_x(0,0) = \lim_{x \rightarrow 0} \frac{u(x,0)-u(0+0)}{x} = \lim_{x \rightarrow 0} \frac{x^3}{x^2 \cdot x} = 1$$

$$\Rightarrow u_y(0,0) = \lim_{y \rightarrow 0} \frac{u(0,y)-u(0+0)}{y} = \lim_{y \rightarrow 0} \frac{-y^3}{y^2 \cdot y} = -1$$

$$\Rightarrow v_x(0,0) = \lim_{x \rightarrow 0} \frac{v(x,0)-v(0+0)}{x} = \lim_{x \rightarrow 0} \frac{x^3}{x^2 \cdot x} = 1$$

$$\Rightarrow v_y(0,0) = \lim_{y \rightarrow 0} \frac{v(0,y)-v(0+0)}{y} = \lim_{y \rightarrow 0} \frac{y^3}{y^2 \cdot y} = 1$$

Thus $u_x = v_y$ and $u_y = -v_x$ i.e. CR equations hold.

Now for Differentiability at origin i.e. $z=0$:

$$f'(0) = \lim_{z \rightarrow 0} \frac{f(z)-f(0)}{z-0} = \lim_{z \rightarrow 0} \frac{\frac{(x^3-y^3)+i(x^3+y^3)}{x^2+y^2} - 0}{x+iy} = \frac{(x^3-y^3)+i(x^3+y^3)}{(x+iy) \cdot (x^2+y^2)}$$

Case I: Take $y = x$

$$f'(0) = \lim_{x \rightarrow 0} \frac{2ix^3}{2x^3(1+i)} = \frac{i}{(1+i)}$$

Case II: Take $y=0$ and $x \rightarrow 0$

$$f'(0) = \lim_{x \rightarrow 0} \frac{x^3(1+i)}{x^3} = (1+i)$$

$\Rightarrow f'(0)$ does not exist. Hence CR equations are true but function is not analytic.

Example: Prove that for the function $f(z) = e^z = e^x(\cos y + i \sin y)$

CR equations are satisfied at $z = 0$ and also function is analytic at origin.

Solution: Given that $f(z) = e^x(\cos y + i \sin y) = e^x \cos y + i e^x \sin y$

Then $u(x, y) = e^x \cos y$ also $v(x, y) = e^x \sin y$

Then at origin

$$\Rightarrow u_x(0,0) = \lim_{x \rightarrow 0} \frac{u(x,0)-u(0+0)}{x} = \lim_{x \rightarrow 0} \frac{e^x-1}{x} = 1$$

$$\Rightarrow u_y(0,0) = \lim_{y \rightarrow 0} \frac{u(0,y)-u(0+0)}{y} = \lim_{y \rightarrow 0} \frac{\cos y-1}{y} = \lim_{y \rightarrow 0} \frac{-\sin y}{1} = 0$$

$$\Rightarrow v_x(0,0) = \lim_{x \rightarrow 0} \frac{v(x,0)-v(0+0)}{x} = \lim_{x \rightarrow 0} \frac{0-0}{x} = 0$$

$$\Rightarrow v_y(0,0) = \lim_{y \rightarrow 0} \frac{v(0,y)-v(0+0)}{y} = \lim_{y \rightarrow 0} \frac{\sin y}{y} = \lim_{y \rightarrow 0} \frac{\cos y}{1} = 1$$

Thus $u_x = v_y$ and $u_y = -v_x$ i.e. CR equations hold.

Now for Differentiability at origin i.e. $z=0$:

$$f'(z) = u_x + i v_x = e^x \cos y + i e^x \sin y = e^x(\cos y + i \sin y)$$

$\Rightarrow f'(0) = |e^x(\cos y + i \sin y)|_0 = 1$ CR equations are satisfied at $z = 0$ and also function is analytic at origin.

Exercise 18: (visit @ Youtube “learning with Usman Hamid”)

1. Prove that for the function $f(z) = |z|^2$ if CR equations are Satisfied but $f'(z)$ does not exists at any non – zero point.
2. Prove that for the function $f(z) = \begin{cases} \frac{\bar{z}^2}{z} & z \neq 0 \\ 0 & z = 0 \end{cases}$ CR equations are satisfied but $f'(0)$ does not exists at any non – zero point.
3. Prove that for the function $f(z) = \sqrt{|xy|}$ is not analytic at the origin Although the CR equations are satisfied at origin.
4. Examine the nature of the function $f(z) = \begin{cases} \frac{x^2y^5(x+iy)}{x^4+y^{10}} & z \neq 0 \\ 0 & z = 0 \end{cases}$ in a region including the origin.
5. Prove that for the function $f(z) = \begin{cases} \frac{xy^2(x+iy)}{x^2+y^4} & z \neq 0 \\ 0 & z = 0 \end{cases}$ is not Analytic at the origin Although the CR equations are satisfied at origin.
6. Prove that for the function $f(z) = \begin{cases} e^{-z^{-4}} & z \neq 0 \\ 0 & z = 0 \end{cases}$ is not analytic at the origin Although the CR equations are satisfied at origin.
7. For the function $f(z) = x^3 + i(1 - y)^3$ Prove that $f'(z)$ Exists only when $z = i$
8. For the following functions show that $f'(z)$ does not exist as CR equations are not satisfied.
 - i. $f(z) = \bar{z}$
 - ii. $f(z) = z - \bar{z}$
 - iii. $f(z) = 2x + ixy^2$
 - iv. $f(z) = e^x e^{-iy}$
9. Show that $f'(z)$ and its derivative $f''(z)$ exist every-where and find $f''(z)$ when
 - (a) $f(z) = iz + 2$
 - (b) $f(z) = e^{-x} e^{-iy}$
 - (c) $f(z) = z^3$
 - (d) $f(z) = \cos x \cosh y - i \sin x \sinh y$.
10. Determine where $f'(z)$ exists and find its value when;
 - (a) $f(z) = 1/z$
 - (b) $f(z) = x^2 + iy^2$
 - (c) $f(z) = z \operatorname{Im} z$.
11. Show that the function $f(z) = x + 4iy$ is nowhere differentiable.
12. Show that the function $f(z) = z^2 + z$ is analytic.
13. Show that the function $f(z) = 2x^2 + y + i(y^2 - x)$ is not analytic.

Example: Prove that the essential characteristic for a function to be analytic

Is that it is a function of 'z' alone, it does not involve $\bar{z}$ or $\frac{\partial f}{\partial \bar{z}} = 0 = f_{\bar{z}}$

Solution: Given that $f(z)$ is analytic. $u_x = v_y$ and $u_y = -v_x$

We know that $z = x + iy$ and $\bar{z} = x - iy$ then $x = \frac{z + \bar{z}}{2}$ also $y = \frac{z - \bar{z}}{2i}$

$$x_{\bar{z}} = \frac{1}{2} \text{ also } y_{\bar{z}} = -\frac{1}{2i}$$

$$\text{Consider } \frac{\partial f}{\partial \bar{z}} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \bar{z}} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \bar{z}} \Rightarrow f_{\bar{z}} = f_x \cdot x_{\bar{z}} + f_y \cdot y_{\bar{z}} = \frac{1}{2} f_x - \frac{1}{2i} f_y \dots\dots\dots(i)$$

$$\text{Now } f_x = u_x + iv_x \text{ also } f_y = u_y + iv_y$$

$$(i) \Rightarrow f_{\bar{z}} = \frac{1}{2} f_x - \frac{1}{2i} f_y = \frac{1}{2} (u_x + iv_x) - \frac{1}{2i} (u_y + iv_y)$$

After using CR equations. i.e. $u_x = v_y$ and $u_y = -v_x$

$$f_{\bar{z}} = \frac{1}{2} u_x + \frac{i}{2} v_x - \frac{i}{2} v_x - \frac{1}{2} u_x = 0 \Rightarrow \frac{\partial f}{\partial \bar{z}} = 0 = f_{\bar{z}}$$

Remark: If a function involve $\bar{z}$ then without verifying CR equations we can Say the function is non - analytic.

For example: $f(z) = \sin(16x + 24iy) = \sin(20z - 4\bar{z})$ involves $\bar{z}$ therefore It is non - analytic.

Exercise 19: (visit @ Youtube “learning with Usman Hamid”)

1) Without verifying CR equations , Prove that following function are non - analytic.

- i. $f(z) = \sin(37x + 35iy)$
- ii. $f(z) = \cos(7x + 5iy)$
- iii. $f(z) = \sin(x - iy)$
- iv. $W = 2ix$
- v. $W = -4y$
- vi. $W = |z|$

2) Show that each of these functions is nowhere analytic:

$$(a) f(z) = xy + iy \quad (b) f(z) = 2xy + i(x^2 - y^2) \quad (c) f(z) = e^y e^{ix}$$

Theorem: If $f'(z) = 0$ everywhere in a domain D, then $f(z)$ must be constant.

Proof: Since $f(z)$ is analytic. Therefore CR equations hold. i.e.

$$u_x = v_y \text{ and } u_y = -v_x$$

now let $f(z) = u + iv \dots\dots\dots(i)$

$$\Rightarrow f'(z) = u_x + iv_x \Rightarrow 0 = u_x + iv_x \Rightarrow u_x = 0, v_x = 0 \Rightarrow u = c_1, v = c_2$$

$$\Rightarrow f(z) = c_1 + ic_2 \Rightarrow f(z) = c_3 = \text{constant}$$